

- ## Technical Feasibility

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|                                  | <ul style="list-style-type: none"> <li>• ANR depends on natural ecological processes rather than intensive planting.</li> <li>• Involves controlled clearing of invasive species to promote native tree regrowth.</li> </ul> <p><b>Environmental Feasibility</b></p> <ul style="list-style-type: none"> <li>• Strengthens slopes by reintroducing deep-rooted vegetation.</li> <li>• Maintains ecological balance by minimizing habitat disruption.</li> <li>• Promotes native species and wildlife habitats..</li> </ul> <p><b>Economic Feasibility</b></p> <ul style="list-style-type: none"> <li>• Eliminates the need for expensive nursery-grown tree plantations.</li> <li>• Sustainable timber, non-timber forest products (NTFPs), and conservation-based tourism.</li> <li>• Reduces expenses related to erosion control and watershed management.</li> </ul> <p><b>Social Feasibility</b></p> <ul style="list-style-type: none"> <li>• Engages local farmers in planning, implementation, and maintenance.</li> <li>• Enhances food security and economic opportunities.</li> <li>• Aligns with government policies and local governance structures for long-term sustainability.</li> </ul> |            |               |             |
| <b>Operation and maintenance</b> | <p>Assisted Natural Regeneration (ANR) in the CHT involves a number of crucial tasks, such as preserving naturally existing seedlings and saplings by controlling invasive species, fires, and grazing. To assess the regeneration process, identify and handle emerging risks, and maintain the wellbeing of the regenerating vegetation, regular surveillance is crucial. In order to maintain the land, soil conservation techniques including mulching and erosion control are used. Furthermore, local communities actively participate in resource management and land protection, which is essential for ensuring sustainable land use practices. To guarantee long-term ecological recovery, ANR requires constant monitoring, assistance, and adaptation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
| <b>Cost</b>                      | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
| <b>Benefits</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                                  | Canopy Cover Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |             |
|                                  | Carbon Sequestration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Crop Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |             |
|                                  | Crop Failure Risk Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
|                                  | Ensuring Food Security                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
|                                  | Fruit Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |               |             |
|                                  | Forestry Goods Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | Livestock Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |               |             |
|                                  | Fisheries Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |               |             |
|                                  | Alternative Livelihood Generation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|                                  | Increase of Income                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | Soil Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Improved Soil Health                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |

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|                                  | Soil Fertility Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                                  | Sustainable Water Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |             |
|                                  | Water Quality Improvement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |               |             |
|                                  | Water Purification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | River Bank Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Surface Runoff Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |             |
|                                  | Groundwater Recharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Pest and Disease Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
|                                  | Reduction of Pesticides and Fertilizer Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Ecosystem and Biodiversity Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |
|                                  | Better Communications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |             |
|                                  | Better Education                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
|                                  | Better WASH facilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
| <b>Implementation Challenges</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                                  | Land unavailable/high land area required                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |             |
|                                  | Supply chain disconnection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |               |             |
|                                  | Land right related problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
|                                  | High human investment capital need                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                                  | Insufficient infrastructures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Less demand of agroforestry product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |
|                                  | Environmentally unsustainable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |               |             |
|                                  | Socially unacceptable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |             |
|                                  | Less beneficial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
| <b>Environmental performance</b> | <ul style="list-style-type: none"> <li>• In the CHT, ANR stabilizes the soil on steep slopes, lessening the possibility of landslides and soil erosion, particularly during periods of intense precipitation. ANR enriches the soil by preserving and encouraging natural vegetation, which increases soil fertility through nutrient cycling and organic matter accumulation.</li> <li>• ANR increases groundwater recharge, decreases surface runoff, and improves soil water retention, all of which contribute to improved water availability during dry periods. Native plant species are restored by the natural regeneration process, which additionally benefits local wildlife habitats and protects the region's abundant biodiversity.</li> <li>• By facilitating the natural regrowth of forests and lessening the stress on the land, ANR offers an alternative to detrimental land-clearing techniques like shifting cultivation. Regenerating forests reduce the carbon footprint of the area and assist in mitigating climate change by sequestering carbon dioxide.</li> <li>• By aiding in the ecosystem's adaptation to shifting weather patterns, the restoration of native plants increases the landscape's overall resilience to the effects of climate change. In order to ensure the long-term health of the forest</li> </ul> |            |               |             |

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|                | ecosystem, ANR encourages sustainable land-use practices that enable local populations to restore damaged land.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Remarks</b> | <p>Effective implementation of Assisted Natural Regeneration depends on careful site selection, focusing on degraded forests with a viable seed bank. Sustained protection is guaranteed only through robust community participation and government support, preventing excessive harvesting. Incorporating livelihood strategies that promote sustainable, forest-based income sources fosters local acceptance and commitment. Finally, a supportive policy and legal framework, underpinned by secure land tenure and incentives for conservation, can significantly boost adoption rates and strengthen long-term outcomes.</p> |